

The Unity of the Wave: Matter, Energy, and Consciousness as Aspects of Frequency

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Abstract

This work presents an extended concept of **wave equilibrium**, proposing that the Universe did not emerge from an explosion or chaos but from a **zero state of harmony** — the balance of oscillations. The foundation of existence is **frequency**, the primary characteristic that defines all forms of matter, energy, and consciousness. Geometry, space, and time arise as consequences of interactions between waves differing in phase and velocity.

The model explains how the structure of the universe emerges from a zero balance: a point is the source of wave-like change. Increasing the base frequency according to the formula: $v_n = (2\pi v)^n$ **where $n > 0$, $n \in \mathbb{Z}$** , forms dimensions – a line, circle/plane, sphere/space, etc. This gives rise to the **fractal organization of the Universe**, where each subsequent dimension, in the geometric sense, is an embedded rotation and resonance of the previous one.

The study distinguishes between the **physical and spiritual** realms as domains of differing frequencies. An increase in frequency leads to spiritual perception; a decrease results in physical density. Matter and consciousness are revealed as two aspects of the same wave, reflecting one another across the spectrum of being.

In its philosophical interpretation, the **centripetal force** that sustains all motion manifests at various levels as gravity, attraction, and love. Here, **love** is understood not as emotion, but as the **fundamental force of balance** — the dynamic that unites all layers of existence.

The human being is described as the **point of intersection** between the physical and the spiritual — a node of equilibrium through which the Universe becomes aware of itself. Thus, the evolution of consciousness is not the accumulation of knowledge, but the **increase in the frequency of awareness**, a movement toward the state where matter, energy, and spirit converge into one.

Keywords:

wave, frequency, equilibrium, consciousness, matter, energy, fractality, spiritual world, physical world, resonance, centripetal force, love, dimensions

1. Introduction

Modern physics and philosophy increasingly face the need to reconsider the fundamental nature of reality. The concepts of space, time, energy, and consciousness — when treated separately — no longer provide a complete description of existence. It is becoming evident that the foundation of being is **a wave process**, in which motion and rest, matter and spirit, cause and effect are not opposites but interdependent states of one unified system.

If we view the Universe not as a collection of objects but as a **single field of oscillations**, a natural question arises: what maintains this field in equilibrium? How can structure exist if everything is in motion? The answer may lie in the nature of the wave itself — in its ability to preserve balance between opposing phases, creating a state of **dynamic equilibrium**, where the sum of all oscillations equals zero.

Thus emerges the idea of **wave equilibrium**: everything that exists is an expression of one principle — the self-regulation of frequencies seeking phase correspondence. Each particle, each field, each act of perception is the result of interacting waves resonating with one another.

From this viewpoint, the **zero state** is not emptiness. It is the ultimate form of stability, where positive and negative energies mutually compensate, while remaining active — generating the pulsation of existence itself.

Zero is not the absence of being, but equilibrium. Not an end, but a beginning — the state where rest and motion are identical.

This work develops the hypothesis that the entire Universe can be understood as a **balanced state of zero**, in which geometry, frequency, and consciousness are three aspects of a single wave structure. This hypothesis seeks to unify the physical and metaphysical levels of description by demonstrating that the law of energy conservation, the perception of form, and the evolution of consciousness share a common source—the **wave process and resonance**.

2. The Nature of Wave Equilibrium

Typically, the origin of the world is associated with a moment of disturbance from a state of rest—with an impulse, an explosion that gave rise to motion, matter, and time. However, such an "explosion" can be viewed as the birth of a new dimension for the wave process—the birth of a new resonant level for a specific frequency of oscillations, a new dimension. As a result, the process gains a new "territory for development," a new level of perception, which immediately establishes the possible parameters for the process's further evolution—this is the Big Bang, the instantaneous speed of space generation.

Thus, an alternative approach is possible: the world could have arisen not from an explosion, but from equilibrium; not from a rupture, but from an internal **balance**, in which opposites do not annihilate each other but enter into interaction.

In this case, the "Big Bang" is not the beginning of everything, but merely a part of a process. Moreover, it is a process directly connected to our Universe and to us. Its origin would be a certain point, which in the physical sense can be represented **as a zero point**—a region where the sum of all energies is zero, but motion does not disappear. In this point, there exists a potential for oscillation: any displacement in one direction causes a reciprocal displacement in

the other. Thus, from zero itself, a wave is born—a sinusoid, the first form in which equilibrium can exist in motion.

The wave is zero that has learned to oscillate.

The sine wave is not merely a mathematical abstraction; it is the universal form through which the **law of energy conservation** manifests itself. Each positive half-wave is counterbalanced by a negative one, and their total remains constant. Motion continues endlessly, but the overall energy remains zero. This is not chaos — it is **the life of equilibrium**.

Therefore, the wave requires no external source. It is self-sustaining: from zero it creates motion, from balance — dynamics, from symmetry — form. The entire Universe can be viewed as a system of interlinked waves, each maintaining equilibrium within its own oscillation.

This understanding changes the perspective on the origin of dimensions, matter, and consciousness. Form, energy, and time are not the products of an explosion, but rather a result generated by the phenomenon of resonance, by the interaction of waves. Physical reality emerges where equilibrium becomes stable, and perception arises where the system continues its development—by increasing its frequency.

The world did not begin with an explosion—it began with the oscillation of zero, with balance.

3. The Sine Wave as the Form of Zero Equilibrium

When the first oscillation arises from zero, it cannot be arbitrary. Any deviation must preserve the overall balance; otherwise, equilibrium collapses and the system ceases to exist. Therefore, the only stable form of motion capable of maintaining equilibrium is the **sine wave**.

The sine wave is both a mathematical and physical expression of **pure balance**. Each of its points is in constant motion, yet the sum of all its states always equals zero. This makes it the perfect form of **dynamic equilibrium**. The positive and negative phases do not oppose but complement one another, creating a continuous flow where energy neither appears nor disappears but simply passes from one half-wave to the next.

In the sine wave, nothing is excessive — every part is balanced by its opposite.

For this reason, the **law of conservation of energy** is not an external constraint imposed upon nature but an internal consequence of form itself. Energy is conserved because its mode of existence is wave-like, and a wave cannot violate its own equilibrium — to do so would mean to cease being a wave.

The sine wave defines the fundamental **geometry of motion**. Within it, all opposites are unified: rise and fall, impulse and return, maximum and minimum. This alternating rhythm is what makes existence possible. If motion lacked a counter-phase, it could not persist — there would be nowhere to return. Zero would become an endpoint rather than a center.

But the sine wave makes zero the **center of the cycle**, the point through which all motion passes, where every half-wave returns to its origin. Thus arises continuity — the foundation of time, form, and energy.

The sine wave is the breathing of zero, the form in which stillness becomes motion and motion sustains stillness.

This form is universal: it manifests in light and sound, in the oscillations of particles, in the heartbeat, in the rhythms of planets and the pulsation of galaxies. All exist only because they internally preserve wave equilibrium — a continual return to zero.

Thus, the sine wave is not merely a shape of oscillation but the **form of existence of the Universe** itself, in which the conservation law arises naturally, and zero is not absence, but the axis of symmetry of all processes.

4. Frequency as the Primary Essence

If the sine wave is the form of equilibrium's existence, then **frequency** is the measure of its manifestation. It is not merely a parameter describing oscillation; it is the **very essence of the process**, defining the structure of time, matter, and perception.

Frequency is the rhythm with which equilibrium oscillates around zero. It determines the scale of perception — what we call “motion” or “rest.” For one system, a slow oscillation appears still; for another, it is instantaneous. Thus arises relativity: the speed of perception depends on the observer's own frequency.

Every state of being can be viewed as a wave of certain frequency, and the interaction between such waves — as a process of resonance. **Resonance** is the natural tendency of frequencies toward harmony, the manifestation of the universal principle of equilibrium — the drive toward phase correspondence, toward a shared rhythm.

In physics, this appears as synchronization and energy conservation. In biology — in the rhythm of the heart, breathing, and cellular cycles. In the psyche — as states of inner harmony and calm. In the spiritual dimension — as the sense of unity with the whole. All these are expressions of one principle: **existence itself is the interaction of frequencies**.

Thus, frequency is not a property of form, but the source from which form arises. It is the fundamental language of nature — through which zero unfolds into multiplicity, and equilibrium becomes motion.

Frequency is the voice of zero, the manifestation of equilibrium in time.

In the work "Hypothesis of Wave Equilibrium: The World as Wave Equilibrium of the Zero State" <https://zenodo.org/records/17386675>, the principle of the emergence of dimensions was described. The essence lies in the formation of connected frequency levels according to the formula:

$$v_n = (2\pi v^2)^n \text{ where } n > 0, n \in \mathbb{Z}$$

Here, the factor 2π participates as the threshold of a new dimension.

4.1. Physical Meaning

In all wave processes, 2π is the completion of a cycle, the moment when the wave returns to its initial phase. But if we consider the Universe not as a flat process, but as a nested structure of frequencies, then each transition through 2π becomes not a return, but a transition to a new frequency level.

$$f_{n+1} = 2\pi f_n$$

Thus, each complete cycle gives birth to a new form—not a repetition of the old one, but a fractal expansion of the previous state. 2π is the universal coefficient of phase transition, through which a new dimension manifests.

4.2. Connection with Geometry and Energy

Geometrically, 2π connects linear motion and rotation: a point, making a full revolution, gives birth to a circle; a circle gives birth to a sphere. In an energetic sense, this multiplier connects frequency and energy through

$$E = h\nu = \hbar\omega = \hbar \cdot 2\pi\nu$$

where $\hbar = h / 2\pi$ is the reduced Planck constant, arising precisely from this principle.

Thus, the factor 2π is not just a mathematical coefficient, but a universal scale of frequency unfolding. It connects mathematics, physics, and the very principle of existence.

4.3. Consciousness and 2π

For consciousness, 2π can be considered as the boundary of awareness. When the frequency of perception reaches the full phase of the cycle, awareness "closes upon itself" and gives birth to a new plane of understanding—a new dimension of perception. This is an act of internal expansion, analogous to the physical transition into a higher space. Therefore, 2π is a bridge between the physical and spiritual dimensions, between the completion of one cycle of experience and the beginning of a new level of awareness.

4.4. Philosophical Conclusion

2π is the frequency law of unity. Each rotation is not an end, but an unfolding into the new. Through 2π , the infinity of the world preserves the equilibrium of zero, manifesting itself ever more deeply—in matter, energy, and consciousness.

5. The Interferential Nature of Perception: The Physical and the Spiritual Worlds

If frequency is the primary essence, then perception is the result of the **interaction of frequencies** — the base and the individual. The base frequency is the universal rhythm of equilibrium, the zero point through which all existence passes. Consciousness is a wave capable of altering its own frequency and entering into resonance with this fundamental harmony.

Depending on their ratio, two directions of manifestation arise — **upward**, toward the spiritual, and **downward**, toward the physical.

When the individual frequency of a subject becomes **higher than the base**, several oscillations appear within one cycle of the base wave — the **birth of subjectivity**, of consciousness as an active phenomenon. The individual frequency begins to perceive several states simultaneously, giving rise to **new dimensions of awareness**. This forms the **spiritual world** — a domain of higher frequency where the process of awareness precedes physical phase.

The spiritual world is a high-frequency dimension where the individual frequency exceeds the base.

If the process goes in the opposite direction — when, within one individual oscillation of an object, several base oscillations are contained — this is a domain of **slowing down**, of materialization. Thus arises the **physical world**, where motion becomes form and phase becomes the structure of matter. The outer world is simply that part of equilibrium where the base frequency dominates over the frequency of the object.

The physical/material world is the domain where the individual frequency is lower than the fundamental wave of existence.

Thus, the spiritual and the physical are not opposites, but **two directions of frequency change**—two modes of existence of the same wave.

The spiritual — is phase acceleration.

The physical — its deceleration.

Reality — their balance.

Life is a mirror in which the balance of the physical and the spiritual is reflected. It sets the **base frequency** — the rhythm in which both sides appear as one.

However, this balance is not static. If consciousness accelerates—increases its frequency—then its physical reflection/perception also changes. The physical world unfolds within the awareness of consciousness.

Subject and object are inseparable. Everything we observe is a reflection of ourselves through the mirror of the fundamental frequency.

This interconnection means that the evolution of consciousness and the evolution of matter are two directions of the same process. Consciousness forms the image of the world, and the world returns information about its state. Thus manifests the universal law of wave equilibrium — everything that changes within finds reflection without.

6. The Mechanism of Dimensional Genesis

If perception is the result of the interaction between the fundamental frequency and the individual frequency, then dimensions are the stages of unfolding this interaction. Each new dimension arises not in space, but in a change of the frequency relationship between the observer and the fundamental wave.

6.1. From Point to Wave

The beginning is a point — equilibrium, zero. There is no motion within it, yet it holds the **potential for oscillation**. The first deviation gives rise to a wave. Its frequency equals the minimal possible rhythm — the base frequency of existence. Thus appears the **first form** — the sine wave, in which positive and negative half-phases balance each other, preserving zero. Here the **law of energy conservation** manifests: the sum of all phases remains equilibrium.

The universe did not emerge from an explosion — it unfolded from balance.

6.2. From Wave to Form

The rotation of a wave around its own axis creates a circle — the first spatial manifestation of form. If this circle is “spun” around the same central point, it produces a sphere — **the second structural dimension**, a volumetric equilibrium in which every surface point lies at an equal distance from the center. This represents complete symmetry — the finished form of that level.

But rotation is not merely movement; it is a **mechanism for maintaining phase**. Each additional rotation introduces a higher frequency, a new axis, and thus a new dimension of perception.

6.3. Frequency as the Generating Factor

Each dimension contains the previous one not only as a projection but also as a frequency component. When the frequencies of two levels enter into a harmonic ratio related to the fundamental frequency—through that very frequency and the 2π multiplier—resonance arises between them: a channel for transmitting energy and information. But this resonance is very distinctive: it allows for the separation of energy exchange processes through quantization, because the higher frequency relative to the lower one again represents a wave, which quantizes the process across different phases. Thus, the spiritual can influence the physical, and the physical can reflect the state of the spiritual. This connection does not require space—it is realized through fractal resonance. And all of this occurs in a quantized manner.

Each new dimension is a new acceleration of frequency by a factor involving the multiplier 2π , giving rise to the perception of a wave relative to the original wave.

6.4. The Nested Nature of Frequencies

Viewed through the principle of fractal resonance, each dimension contains the previous ones, just as higher frequencies include lower harmonics. This explains the interconnection of all levels of reality: the physical world is the low-frequency manifestation, the spiritual — the high-frequency domain, both belonging to one and the same wave — the unified equilibrium of zero.

6.5. Human Perception

Dimensions are not born from geometry but from the acceleration of frequency. When the frequency of a system increases by a multiple of itself, it begins to perceive the previous oscillation as a point. This is the birth of fractality — the emergence of a new dimension. Thus, consciousness “emerges” from the point (time) into a plane, from the plane into a volume, and from the volume into a level that is beyond our comprehension, as it requires a frequency higher than our capacity of perception.

We do not live *in* a three-dimensional world — we *are* a three-dimensional perception of a higher reality.

7. Fractal Interconnection of Dimensions. The Resonant Structure of Reality

If each dimension represents a new degree of phase acceleration, then there are no real gaps between them. They are not isolated but linked through the **laws of resonance**. Reality is not a collection of separate levels, but a **unified field of nested frequencies**, where each structure is a reflection of all others within its own phase.

7.1. Resonance as the Principle of Connection

Each dimension contains the previous one not only as a projection but also as a frequency component. When the frequencies of two levels come into a multiple relationship connected through the base frequency itself, a resonance arises between them — a channel for the transmission of energy and information. Thus, the spiritual can influence the physical, and the physical can reflect the state of the spiritual. This connection does not require space — it is realized through fractal resonance.

Resonance is the language through which dimensions communicate with one another.

In this way, the mutual influence of all levels is formed: any change at one level creates a response throughout the entire system. Consciousness and matter turn out to be inseparable, for they belong to the same wave, differing only in the frequency range of perception.

7.2. Fractal Structure

The nesting of dimensions forms the **fractal organization of the Universe**. Each level is a smaller or larger replica of another, but with its own frequency and scale of manifestation. Thus, **elementary particles repeat the structure of galaxies**, and **galactic structures mirror the patterns of the microcosm**. This self-similarity is not accidental — it is a consequence of **wave equilibrium**.

When reality is viewed as a wave, fractal organization becomes the natural state of stability. Every whole seeks to preserve balance by reproducing itself at different frequency levels. This is not the repetition of form, but the repetition of the **principle of equilibrium**.

Fractality is the form that equilibrium takes when it manifests in time.

7.3. Dimensions as Harmonics

Each dimension can be considered a fractal harmonic of the fundamental frequency (an increase by a frequency factor and the 2π multiplier—a wave upon a wave). The first dimension is the fundamental tone, the second is its first fractal harmonic, and so on. In this way, the entire Universe becomes a musical structure, where every form resonates at its own frequency, yet all together compose a single chord of existence.

This harmonic organization explains the law of energy conservation and the informational continuity of the world. Energy is not lost—it merely changes its frequency range, transitioning from the dense to the subtle, from the material to the spiritual, from form to awareness.

7.4. Consciousness

Consciousness occupies a unique position — it is capable of awareness. When consciousness increases its frequency, it perceives new levels of reality; when it decreases, those levels fade from its field of perception.

8. The Law of Conservation as a Manifestation of Equilibrium. Energy as the Form of Connection Between Worlds

If reality is a system of interconnected frequencies, then energy is the measure of their interaction. It belongs neither to matter nor to spirit — it is the link between them, a form of equilibrium that arises through the exchange of phase.

Energy does not exist by itself; it is the result of phase difference and resonance between levels. When two waves coincide in frequency and phase, resonance becomes complete, and energy is perceived as stillness. When a phase shift occurs, motion appears — the flow we perceive as time, heat, or light.

Moreover, energy can move from level to level, from dimension to dimension. Dimensions become the elements that produce the quantization of energy.

Energy is the form through which equilibrium reveals its disturbance.

8.1. Conservation as Dynamic Equilibrium

The law of energy conservation does not imply immobility. It states that the sum of all processes always equals zero, because every positive phase is balanced by a negative one. This is not static but **dynamic symmetry**, in which oscillation does not disappear — it only redistributes across levels.

Physics perceives this as energy transfer between bodies. Consciousness experiences the same process — as a shift in states, emotions, and insights. In both, what remains constant is not matter, but the **balance of interaction**.

Conservation is the memory of equilibrium.

8.2. Energy as a Bridge Between Worlds

The base frequency defines zero — absolute balance. The physical world is the domain where energy takes form, and the spiritual world — where it becomes awareness. Energy connects these two directions, allowing equilibrium to exchange information through frequency modulation.

When consciousness accelerates, a portion of energy transforms into subtle forms — intuition, inspiration, the sense of unity. When the frequency decreases, the same energy condenses — becoming motion, mass, or heat. The same process appears as matter or thought, depending on the observer's frequency range.

Energy is the language through which spirit speaks to matter.

8.3. Equilibrium as the Source of All Laws

All physical laws are partial expressions of the principle of equilibrium. The law of conservation, symmetry, entropy, and action–reaction — all describe the same thing: the world's tendency to preserve zero. Equilibrium is not a state of rest, but the **essence of being itself**, through which motion, form, and awareness become possible.

Consciousness, as part of this process, does not merely observe equilibrium — it **restores it** within itself, linking the spiritual and physical through conscious interaction.

Energy is the breath of equilibrium. Consciousness — its awareness.

9. The Trinity of the Wave: The Unity of Matter, Energy, and Consciousness

Everything that exists is a manifestation of the wave — a process that unites **three inseparable aspects**: matter, energy, and consciousness. They are not independent entities but reflections of one another, just as amplitude, frequency, and phase are in oscillation. Matter is form, energy is motion, and consciousness is awareness of the phase. Together they form **a single field of equilibrium**, where manifestation becomes possible only through their interaction.

9.1. Matter as Manifested Form

Matter is the phase frozen in time. It carries within itself the trace of all interactions but does not perceive them, since its frequency is below the base frequency. For matter, time is not a direction but an internal rhythm. It moves **within** time rather than through it, realizing a stable state of balance in manifested form. Thus, structure arises — the visible side of the wave.

Matter is the memory of equilibrium imprinted into form.

9.2. Energy as Process

Energy is a transition — an interaction between phases and dimensions. It belongs to neither side but connects them, ensuring continuity. Through energy, equilibrium “breathes” — expanding and contracting, creating the pulse of the universe. Energy is what makes motion possible, while maintaining the zero sum of all changes.

Energy is the breath of equilibrium.

9.3. Consciousness as the Observing Phase

Consciousness is a wave that is aware of itself. It perceives the difference between phases and dimensions, and it is from this difference that perception arises. Consciousness does not observe matter, but the interference — between itself and the world, between the inner and the outer. When the frequency of consciousness aligns with the base frequency, resonance occurs — a state of unity, calm, and clarity. Consciousness becomes transparent to the very process of existence.

Consciousness is equilibrium aware of itself.

9.4. The Wave as a Universal Principle

At all levels — from the microcosm to galaxies, from physics to psychology — the wave remains the fundamental form of being. It describes both the motion of matter and the oscillation of thought. The wave unites causality and perception; it is the only structure in which the **law of conservation** and the **law of cognition** coincide.

The wave is the universal language of equilibrium.

9.5. The Trinity

Matter, energy, and consciousness are not three separate worlds but three faces of a single essence. Matter embodies energy, energy links them through phase, and consciousness realizes the process itself. They form a **trinity of existence**, in which all opposites dissolve into unity.

When matter becomes aware — it becomes spirit. When consciousness manifests — it becomes matter. Between them lies energy — the eternal mediator of balance.

Thus arises the universal trinity — form, motion, and awareness — reflecting the three fundamental laws of being: **conservation, interaction, and development**. And at the root of them all lies **zero** — the stillness from which everything unfolds.

10. Philosophical Interpretation and Direction of Development

If we view the Universe as a system of waves in equilibrium, it becomes clear that motion, evolution, and awareness are not separate phenomena, but three expressions of one underlying principle — **the evolution of frequency**.

10.1. Frequency as the Path of Conscious Development

Consciousness is a wave capable of changing its own frequency. When its frequency coincides with the base frequency, a state of stillness arises — harmony with the world. When consciousness accelerates, it begins to perceive a greater number of dimensions — an expansion of perception that allows it to become aware even of itself.

Awareness is not knowledge, but the speed with which spirit perceives itself.

10.2. The Centripetal Force as a Universal Principle

At all levels of existence, there is one force — **centripetal**. It maintains the equilibrium of all rotations and frequency states. This force does not destroy; it connects. It unites everything that exists into a single whole. On the physical level it appears as gravity, on the energetic — as attraction, and on the spiritual — as **love**.

Love is not emotion but the **force of equilibrium** that holds all waves in coherent motion. It is both the geometric and spiritual axis of the Universe — the centripetal essence through which all things return to unity.

Love is the centripetal force of spirit.

10.3. The Unity of the Physical and the Spiritual

The physical and spiritual worlds are not opposites, but reflections of the same wave. Physics is the manifestation in the realm of low frequencies, where phase becomes form. Spirit is the realm of high frequencies, where form transforms into awareness. They are connected by the law of resonance—any change in one is reflected in the other. When consciousness accelerates, matter responds by restructuring itself, and when matter condenses, consciousness likewise reacts to maintain balance.

The world is a mirror reflecting the frequency of our awareness.

10.4. The Human as a Point of Equilibrium

The human being is the node where the physical and the spiritual intersect. Within, both realities coexist — as form and as awareness. The task of humanity is not to break the balance, but to maintain it consciously — to see the inner and the outer as two poles of one phase. Each act of awareness is not an escape outward, but a movement inward — toward the center.

When thought increases in frequency, it draws closer to the still point of equilibrium, where all opposites dissolve and only pure presence remains — **consciousness aware of itself**.

Everything that exists moves toward equilibrium.

Everything that perceives moves toward unity.

Everything that loves returns to the Source.

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